Yonwoo Choi

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Education

Seoul National University *Mar 2022 – Feb 2024*
M.S in Computer Science and Engineering

Advisor: Hanbyul Joo

Thesis: A Wrist Mounted Camera System for Hand Pose Estimation from Single RGB Images

GPA: 3.73/4.3

Coursework: 3D Computer Vision, Multimodal Deep Learning, Advanced Graphics, Advanced Data Mining, Mobile Computing

University of Virginia *Aug 2019 – May 2021*
B.A in Computer Science

GPA: 3.77/4.0

Coursework: Data Structures, Algorithms, Computer Architecture, Web Applications, Software Testing, Human Computer Interaction

Research Interest

My research lies at the intersection of 3D computer vision and generative AI, with a focus on creating realistic synthetic data for computer vision applications. My work explores the use of generative models to create high-quality synthetic training data, bridging the gap between simulation and real-world performance. I am passionate about developing practical systems that can capture, understand, and reconstruct complex 3D scenes and objects.

Publications

SVAD: From Single Image to 3D Avatar via Synthetic Data Generation with Video Diffusion and Data Augmentation **CVPR 2025 Workshop**

Yonwoo Choi

A Wrist Mounted Camera System for Hand Pose Estimation from Single RGB Images **M.S Thesis**

Yonwoo Choi, Hanbyul Joo

Work Experience

GenGenAI *Seoul, South Korea*
Algorithm Engineer *Sept 2025 – Present*

Generating Synthetic Data with 3D

Controllable Image and Video Generation Across Domains

SECERN AI *Seoul, South Korea*
Research Engineer *Mar 2024 – Sept 2025*

Animatable 3D Human Avatars

Multiview 3D Reconstruction

Visual Computing Lab *Seoul, South Korea*
Graduate Researcher *Mar 2022 – Feb 2024*

3D Egocentric Hand Pose Estimation

Multiview Capture System for Human Pose Estimation

Open Source

Single Image to 3D Human Avatar *May 2025*
https://github.com/yc4ny/SVAD

Official Implementation of the paper SVAD

Synthetic Data Generation Pipeline for 3DGS Avatar Training

Facebuilder *Mar 2025*
https://github.com/yc4ny/facebuilder

3D Face Mesh Deformation with Pin-Based Interaction

Multiview Face Modeling with Real-time 3D Consistency

SMPL Fitting *Jun 2024*
https://github.com/yc4ny/SMPL-Fitting

Fitting SMPL Parameters to 3D Human Joints

Parametric Human Shape and Pose Optimization

Egocentric Human Pose Estimator *Dec 2023*
https://github.com/yc4ny/xR-EgoPose

Unofficial Implementation of the Paper xR-EgoPose

Egocentric 3D Human Pose Estimation from a Single Fisheye Camera

Multiview Human Pose Capture System *Apr 2022*
https://github.com/yc4ny/Multi-Camera-3D-Human-Reconstruction

5 GoPro Camera Setup for 3D Human Reconstruction

Camera Calibration with Zhang's Method

Projects

Missile-Perspective Ground Scene Video Generation *Mar 2026 - Present*

Missile-perspective synthetic video generation for ground vehicle detection.

GenGenAI, Hanwha Aerospace

Missile-Perspective Naval Scene Video Generation *Dec 2025 - Present*

Missile-perspective synthetic video generation for naval target detection.

GenGenAI, Hanwha Aerospace

Multiview 3D X-Ray Detection for Airport Screening *Mar 2024 - Sept 2025*

3D X-Ray Calibration, 3D Bounding Box Detection

SECERN AI, Ministry of Land, Infrastructure and Transport

Teaching

Teaching Assistant, Seoul National University *Fall 2023*

4190.408 001 - Artificial Intelligence (Instructor: Hanbyul Joo)

Teaching Assistant, Seoul National University *Spring 2022*

4190.773 002 - 3D Computer Vision (Instructor: Hanbyul Joo)